

Project Title: Type Word Worm

1. Overview: This project is a cli-based typing game implemented in F# and .NET 10. English words appear from the right and move toward the left side of the screen. Player can get the score and kill the worm by typing the word accurately before they reach the left boundary.

2. Requirements.

The following requirements are designed to be objective and testable.

1. Dynamic Word Generation:

- Words are spawned at the right boundary ( $x = 80$ ).
- The spawn rate starts at 1 words per second and increases by 0.5 every 10 seconds.
- $Spawn\ Rate = 1 + \frac{Time}{10} \times 0.5$

2. Movement and Pathing:

- Words move from right to left with a constant horizontal velocity  $v$ .
- $v = 5 + \frac{Time}{10} \times 0.5$
- The vertical position is randomly decided when word spawns.

3. Scoring and Particle Effects:

- Arrow Projectile: Upon correctly typing all characters of a target word, an arrow projectile (---->) is spawned at the left boundary ( $x = 5$ ). The projectile travels rightward at a constant velocity of 2 times of  $v$ (speed of the word) until it reaches the target word.
- Word Removal: The target word is only removed from the screen when the arrow projectile makes contact with it, triggering a brief circular particle effect
- Score Calculation: Points ( $\Delta S$ ) are awarded at the moment of elimination based on the formula:  $\Delta S = L \times R$ , where  $L$  is the number of characters

in the word and  $R$  is the Current Spawn Rate

#### 4. Termination Condition (Game Over):

- A "Dead Line" is established at the left boundary (  $x = 5$  ).
- If the x-coordinate of any character in a word becomes less than or equal to 5, the game immediately terminates.

#### 3. Example Interaction

System: A word coding spawns at  $x = 80$  and moves leftward in a wave pattern.

User: Types c, o, d, i, n, g (enter).

System: The word coding disappears with a circular particle effect, and the score increases by 10.

System: As time passes, new words spawn more frequently.

System: A word fsharp reaches  $x = 5$ . The game stops and shows a slanted GAME OVER screen.

#### 4. Technical Specifications

- Language: F#
- Runtime: .NET 10
- UI Mode: CLI